AARADHYA DEV TAMRAKAR

Kathmandu, Bagmati 44600 | aaradhyadevtmr@gmail.com | +977 9844602050
linkedin.com/in/aaradhya-dev-tamrakar | github.com/AaradhyaDT | aaradhyadtmr.github.io

SUMMARY

Electronics, Communication and Information Engineering student at IOE, Kathmandu Engineering College (Year IV/Part I). Focused on Artificial Intelligence, robotics, and embedded systems. Experienced in building machine learning pipelines using Python and scikit-learn, deploying models via FastAPI, and developing Android applications with Kotlin/Jetpack Compose. Currently a Fellow at the Fusemachines AI Fellowship (2026) and NSSR DataCamp Fellow (Cohort 2, 2026). Interested in applying AI to real-world sensor systems, robotics control, and edge intelligence.

EDUCATION

Bachelor of Engineering (B.E.) — Electronics, Communication and Information Engineering
Institute of Engineering (IOE), Kathmandu Engineering College *Expected: January 2027 | Year IV / Part I*

Relevant Coursework: Machine Learning · Signals and Systems · Digital Electronics · Microprocessors · Control Systems · Communication Systems · DBMS · Project Management · Propagation & Antenna

EXPERIENCE

Vice Chair *2026–Present*
IEEE KEC Student Branch

- Spearheading strategic planning and high-level decision-making to scale technical initiatives and member engagement.
- Directing cross-functional executive committees and managing institutional relations to enhance the engineering community ecosystem.

Event Manager *2026–2027*
Electronics Project Club (EPC), Kathmandu Engineering College

- Managing logistics and coordination for engineering events and technical workshops.

Vice Secretary *2025–2026*
IEEE KEC Student Branch, Kathmandu Engineering College

- Streamlined branch operations by managing official documentation, meeting minutes, and internal communication frameworks.
- Orchestrated logistics for technical workshops and IEEE-related programs, significantly increasing student participation.
- Supported the executive committee in coordinating volunteers and event participants to ensure seamless event execution.

Resource Manager *2024–2026*
Electronics Project Club (EPC), Kathmandu Engineering College

- Managed logistics and scheduling for engineering workshops and technical events.
- Coordinated resource allocation for projects across multiple departments to optimize utilization and meet deadlines.

FELLOWSHIPS & RECOGNITIONS

Fuse AI Fellow — Fusemachines AI Fellowship *2026 (Active)*
Fusemachines

- Admitted to the Fusemachines AI Fellowship program, a selective AI/ML fellowship in Nepal.
- Passed entrance examination (March 2026) covering linear algebra, calculus, probability, Python, and ML.
- Week 1: Completed heart attack risk prediction pipeline — data wrangling, BP parsing, two-step merge → 8,763 × 22 clean dataset, 0 nulls.
- Positioned at the intersection of AI, education, and hands-on technical building.

NSSR DataCamp Fellow — Cohort 2 *2026 (Active)*
Nepalese Society of Student Researchers

- Selected April 28, 2026; sponsored DataCamp Premium license for structured AI/ML coursework.
- SQL progress: Data Manipulation in SQL Ch.1 (CASE WHEN) complete; Ch.2 (Subqueries) in progress.

- Course priority: Data Manipulation in SQL → PostgreSQL Summary Stats → Pandas → Statistical Thinking → ML with scikit-learn.

PROJECTS

Edge AI Stability Detection System

Python · Scikit-learn · FastAPI · Joblib

- Developed an ML system to predict platform stability using simulated IMU sensor data (tilt_x, tilt_y, angular velocity).
- Generated a 10,000-sample synthetic dataset; trained a Random Forest classifier achieving 99.8% test accuracy.
- Built an end-to-end ML pipeline: data generation → feature engineering → model training → evaluation → deployment.
- Deployed model as a REST API via FastAPI for real-time predictions; exported using Joblib for robotics integration.

GitHub: https://github.com/AaradhyaDT/stability-ai-system

Gesture-Controlled Self-Balancing Robot (GCSBR) — Minor Project

Computer Vision · Arduino · MPU6050 · Stepper Motors · Android · MATLAB

- Examiner described it as 'major project level' at evaluation.
- Developed a computer vision–based gesture recognition system for robotic navigation using MediaPipe landmarks.
- Firmware V6.9.1: watchdog timer, atomic writes, differential ramp, integrator bleed, PID stabilization.
- Built Android gesture-control interface (MediaPipe, CameraX, HC-05 BT); MATLAB simulation → hardware deployment pipeline.
- Implemented dual-hand gesture control for direction and speed; DRV8825 drivers on CNC Shield for real-time actuation.

GitHub: https://github.com/AaradhyaDT/Gesture-Controlled-Self-Balancing-Robot

Alpha Android Super-App

Kotlin · Jetpack Compose · Material3 · DataStore · Apache POI

- Building a modular Android super-app (SDK 36) integrating SBR gesture control, multi-mode calculator, budget tracker with eSewa parser, and Google Drive backup.
- Architected with clean module separation; custom Astronomus font; full Material3 design system.

SysOptimizer — Windows Optimization Tool

Python · CustomTkinter · PyInstaller · WMI · PowerShell

- Built a standalone Windows system optimization tool packaged as .exe; features animated ring gauges, sparklines, process table.
- v4/v5: Power Plan switcher, RAM flush, Background Bloat panel (14 services/tasks), Startup Scanner, per-core CPU reporting.
- Implemented silent background operation via CREATE_NO_WINDOW and -WindowStyle Hidden for zero UI interruption.

SKILLS

Programming: Python, C, C++, Kotlin, SQL
Machine Learning: scikit-learn, NumPy, Pandas, Random Forest, Classification, Model Evaluation
Android Development: Kotlin, Jetpack Compose, Material3, DataStore, CameraX, MediaPipe, Apache POI
Deployment: FastAPI, REST APIs, Joblib
Embedded Systems: Arduino, MPU6050, CNC Shield, DRV8825, Stepper Motors, HC-05 BT, UART
Productivity & Office: Microsoft Office (Word, Excel, PowerPoint), Google Workspace (Docs, Sheets, Slides)
Design & Media: Canva, Clipchamp, Webflow, Blender (Beginner)
Tools: Git, GitHub, VS Code, Jupyter Notebook, Google Colab, MATLAB, SSMS, Google Cloud Console (familiar)
AI / Local LLM: Ollama, AnythingLLM, ipex-llm, DeepSeek R1 7B (Intel Arc iGPU)
Other: Project Management, Event Coordination, Public Relations, Adaptability

TECHNICAL ACTIVITIES & CERTIFICATIONS

- IEEE Conference Leadership Workshop 2026 — IEEE Nepal Section
- IEEE Xtreme 19.0 Participant
- AWS Cloud Computing Workshop — KEC IT Club (2025)
- PreXtreme Competitive Programming Workshop — IEEE Pulchowk (2025)
- Git & GitHub Workshop — KEC IT Club (2024)
- Active member: IEEE KEC Student Branch, Electronics Project Club, KEC Maker's Space, KEC Music Club